

Math 521, Section 2
Midterm 2
Spring 2026

First name: _____ **Last name:** _____

Instructions:

- This exam contains 4 problems, and there are a total of 40 points available.
- Show all your work in the space provided. You may also use the backs of pages.
- No outside resources are allowed, including notes, calculators, textbooks, etc.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
Total:	40	

1. (10 points) Let $\{a_n\}_{n \geq 1}$ be a sequence of positive real numbers. Prove that $\lim_{n \rightarrow \infty} a_n = \infty$ if and only if $\lim_{n \rightarrow \infty} \frac{1}{a_n} = 0$.

2. (10 points) Let $\sum_{n=1}^{\infty} a_n$ be a series of real numbers. Prove that if $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_{2n}$ converges.

3. (10 points) Let (X, d) be a metric space and let A be a nonempty subset of X . Prove that A is open in (X, d) if and only if there exists a set I , positive numbers $r_i > 0$, and points $a_i \in X$ so that $A = \bigcup_{i \in I} B_{r_i}(a_i)$.

4. Recall that $d_2(x, y) = \sqrt{|x_1 - y_1|^2 + |x_2 - y_2|^2}$ is a metric on $\mathbb{R}^2$. Consider the subspace

$$Y = \{(x, y) \in \mathbb{R}^2 : y > 0\}$$

with the induced metric d_Y . **Prove or disprove** the following statements:

- (a) (5 points) If $A \subseteq Y$ is open in (Y, d_Y) , then A is open in $(\mathbb{R}^2, d_2)$.
- (b) (5 points) If $B \subseteq Y$ is closed in (Y, d_Y) , then B is closed in $(\mathbb{R}^2, d_2)$.

Extra paper